Hydro-Reactive Fuel Water Ramjet Propulsion

Final-year major project (B.Tech Aerospace Engineering, MIT-ADT University, Pune) evaluating hydro-reactive metal fuels for water-breathing ramjet propulsion — a candidate technology for air-independent underwater propulsion (torpedoes, high-speed submarines, underwater vehicles).

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Overview

Water-breathing (air-independent) propulsion systems draw their oxidizer from ambient seawater instead of carrying it onboard, using hydro-reactive fuels that release energy on contact with water. This project compares five fuel candidates — pure Aluminum, pure Magnesium, pure Boron, an Al-Mg combination, and Sodium Borohydride ($NaBH_4$) — to identify the most efficient option for a water-ramjet system.

Methodology

- **Thermochemical analysis** of each fuel composition using **NASA CEA** (Chemical Equilibrium with Applications), evaluating specific impulse, chamber temperature, and combustion species across a range of chamber pressures and water-to-fuel ratios.
- **Literature-informed parameters**: reaction efficiencies, metal-loading ratios, and experimental setup details were cross-referenced against prior published water-ramjet studies to ground the theoretical analysis in realistic operating conditions.
- Comparative evaluation of plain fuel + water systems vs. Teflon (C_2F_4)-blended systems, which showed significantly higher flame temperatures and specific impulse.

Key Results

Fuel System	Chamber Temp	Specific Impulse
B + H_2O	~2000–2100 K	—
Mg + H_2O	~2300–2400 K	~900 m/s
Al + H_2O	~2200–2500 K	~900 m/s
B + Teflon	>3500 K	~1600 m/s
Mg + Teflon	~3100 K	~1000 m/s
Al + Teflon	~3200–3400 K	>1000 s

Mg + H_2O emerged as the most practically attractive plain-water system, balancing reaction completeness and performance; Teflon-blended systems showed the highest theoretical performance but at the cost of fluorine-bearing exhaust species.

Tools Used

- NASA CEA (thermochemical equilibrium analysis)
- ANSYS Fluent (CFD, for follow-on computational validation)

Repository Contents

- `/report` — Full project report (PDF)
- `/cea-data` — NASA CEA input/output files for each fuel composition
- `/figures` — Result plots and comparison charts

Notes

This repository contains only material cleared for public sharing. Experimental facility details and data are drawn from published literature cited in the full report.